

Introduction

The primary objective of this analysis is to explore key predictors of team performance in baseball by examining batting records from 1998 to 2024 (excluding 2020) of the Minnesota Twins baseball team. Through exploratory data analysis (EDA) and correlation analysis, patterns and relationships among various metrics such as runs, batting average, runs batted in and hits are identified to better understand their associations with team wins. Multiple regression models are then tested to pinpoint the variables that best predict a team's success, ultimately resulting in a parsimonious two-variable model. This model utilizes runs and batting average as the primary predictors to estimate the number of team wins.

Performance in baseball, often quantified by the number of wins, is influenced by a variety of team and individual statistics. Analyzing historical data allows us to identify patterns and relationships within this complex array of metrics, providing a data-driven approach to predict outcomes. The chosen dataset spans 26 years, offering a comprehensive overview of team performance across seasons, excluding 2020 due to irregularities in the season format. The independent variables examined include common performance indicators such as runs, hits, batting average, home runs, and others, each of which may have varying degrees of association with team wins.

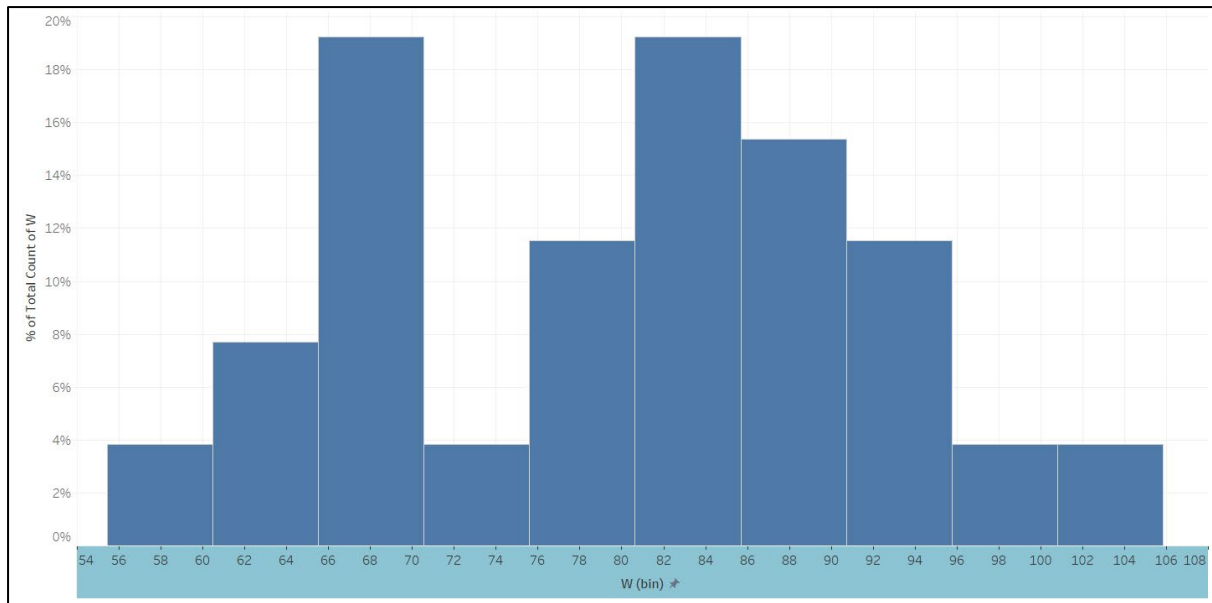
Through exploratory data analysis and correlation analysis, we assess each variable's potential contribution to team success, followed by multiple linear regression to build predictive models. By comparing models with different combinations of variables, we determine that a model based on runs and batting average provides the best balance of predictive power and simplicity. The findings offer insights into the statistical factors most strongly associated with team wins and establish a foundation for further research into baseball performance metrics.

Exploratory Data Analysis

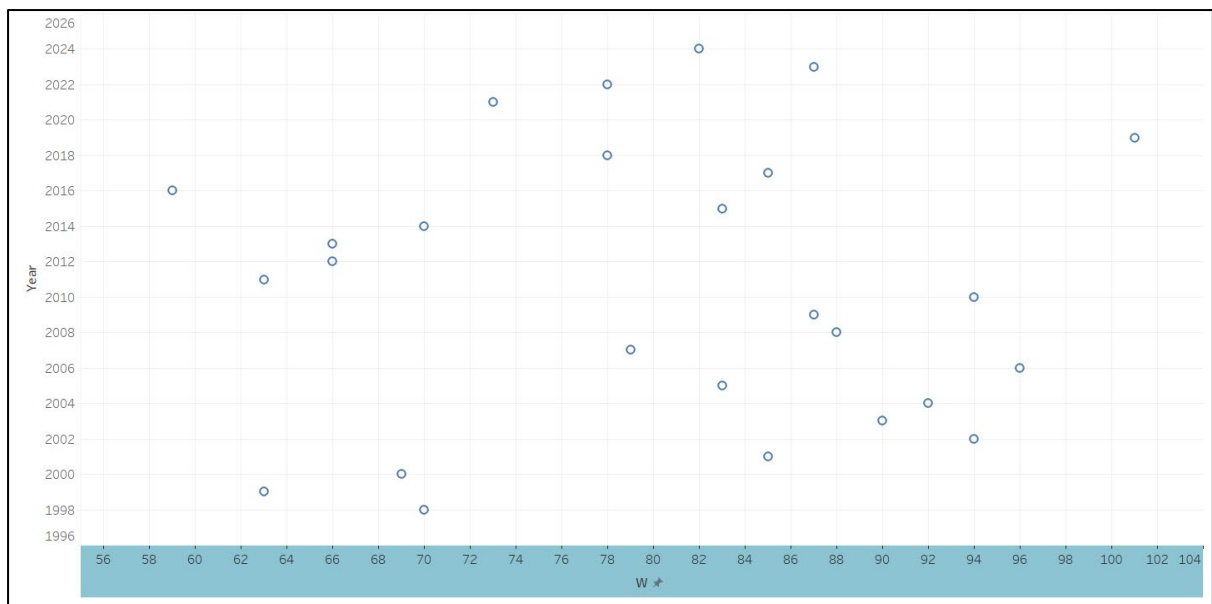
The first step involves analyzing the datasets to summarize their main characteristics through visual methods, allowing us to observe patterns, trends, relationships and identify potential outliers. I will be analyzing the data of batting records of baseball from 1998 to 2024, excluding 2020 i.e., 26 years. Graphs such as histograms and scatterplots will be used to visualize the data.

The potential independent variables I am analyzing are the number of wins, runs, hits, home runs, runs batted in, strikeouts, triples, doubles, walks, batting average, age of batters, and stolen bases. The selected independent variables for this analysis capture key aspects of team performance that influence game outcomes. Offensive metrics like runs, hits, and RBIs reflect a team's scoring capability, while batting average and strikeouts indicate efficiency in hitting. Variables like stolen bases and triples highlight aggressive play strategies that can create scoring opportunities. Additionally, player age provides insights into experience, potentially impacting performance. Together, these variables offer a well-rounded view of factors that are likely to contribute to a team's success and allow for meaningful exploration of performance trends.

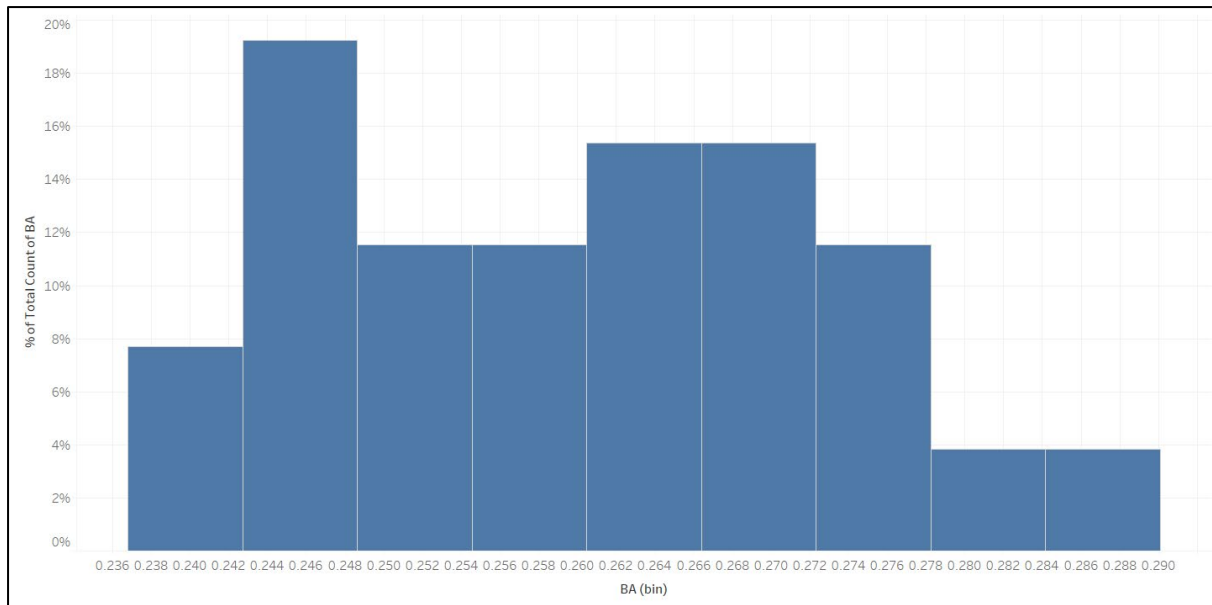
The distribution of the number of wins seems to be bi-modal and there no potential outliers. 19.23% of wins are between 66 and 71 and the other 19.23% wins are between 81 and 86.



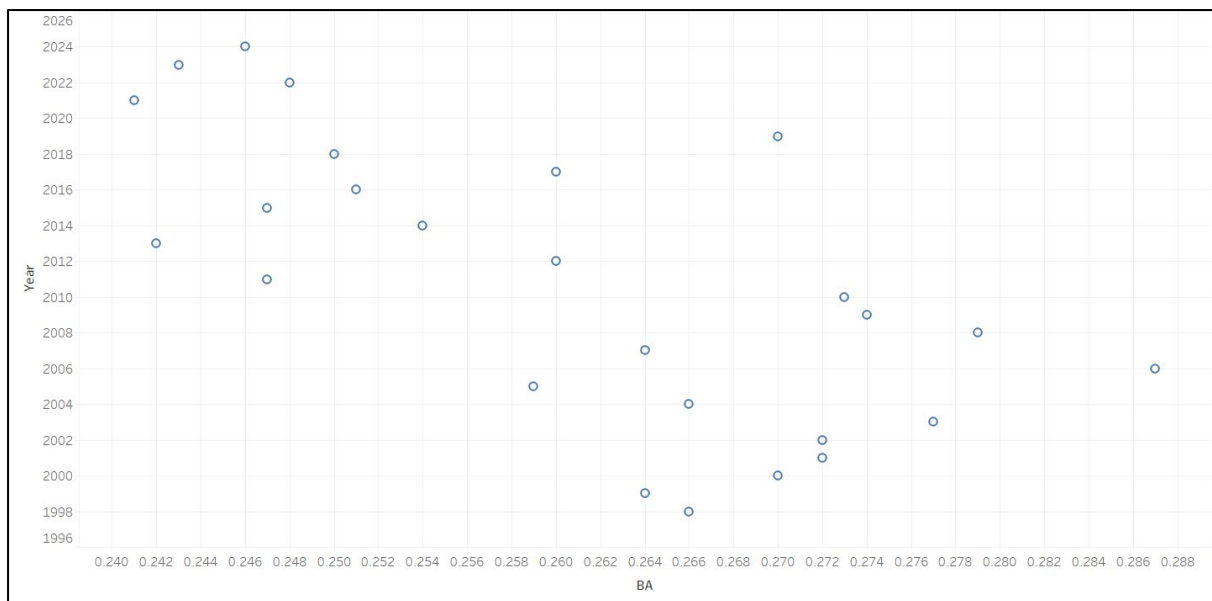
The lowest number of wins are 59 and the highest number are 101. The highest number of wins are in 2019 and lowest in 2016.



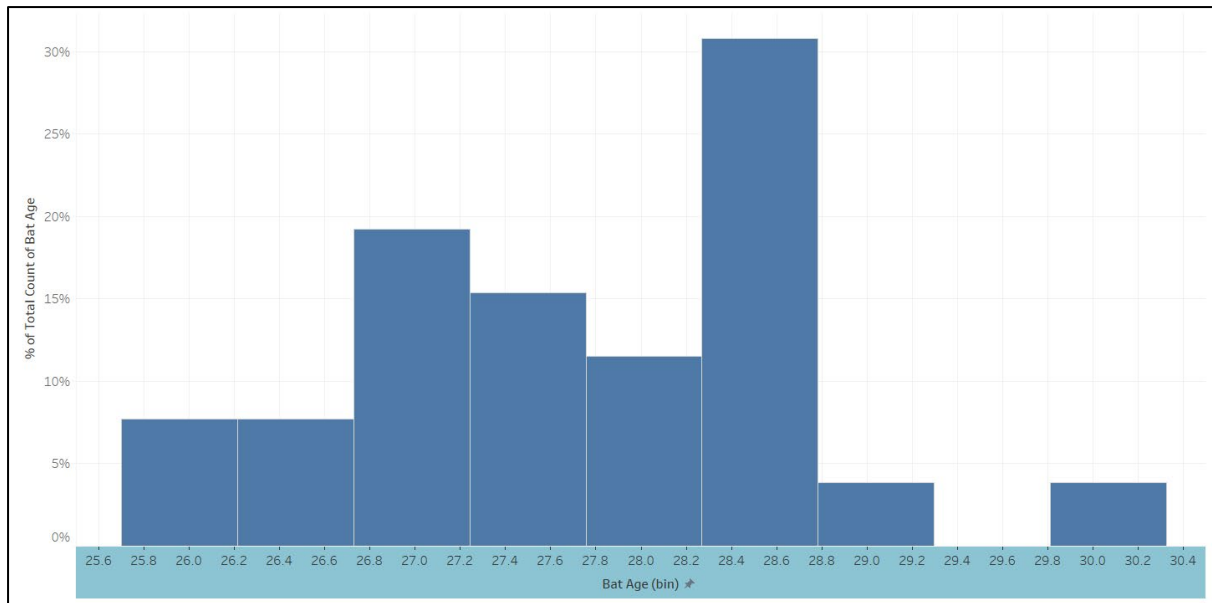
From the distribution of batting average, 19.23% of the total batting average lie between 0.243 and 0.248, there are no potential outliers.



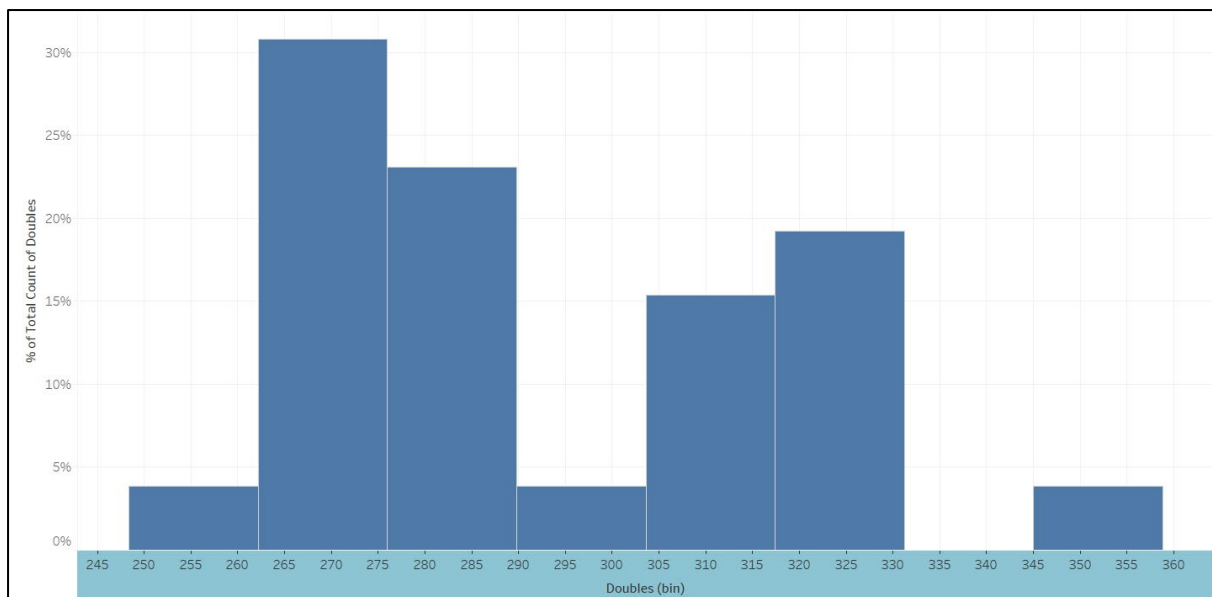
The highest batting average was 0.287 in year 2006 and the lowest was 0.241 in year 2021.



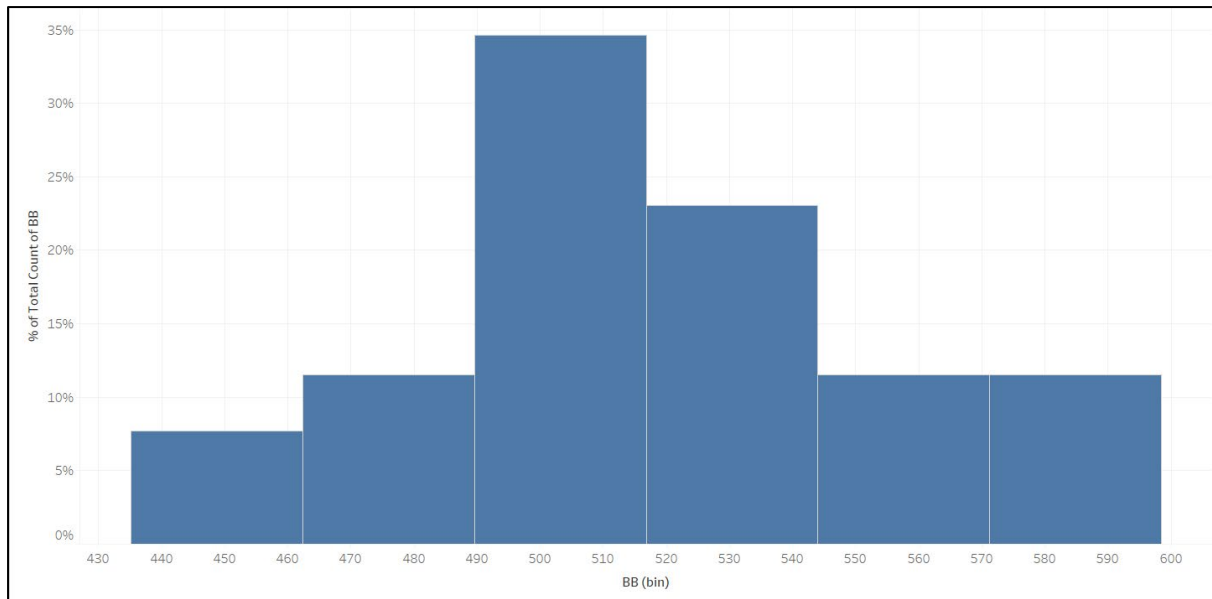
30.77% of the total players are 28.3 to 28.8 years old. There is one potential outlier in the age distribution. The age of the oldest player is 29.8 years, and the youngest player is 25.7 years old.



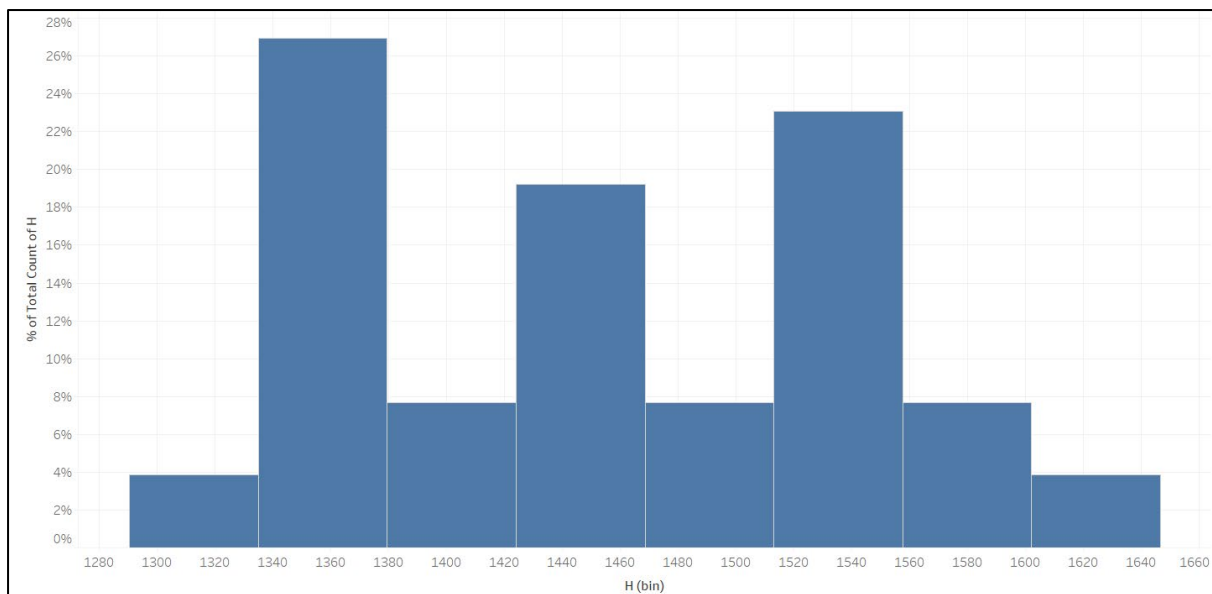
30.77% of total doubles lie between 262 and 275. There is one potential outlier in the distribution. The distribution seems to be skewed a bit to the right.



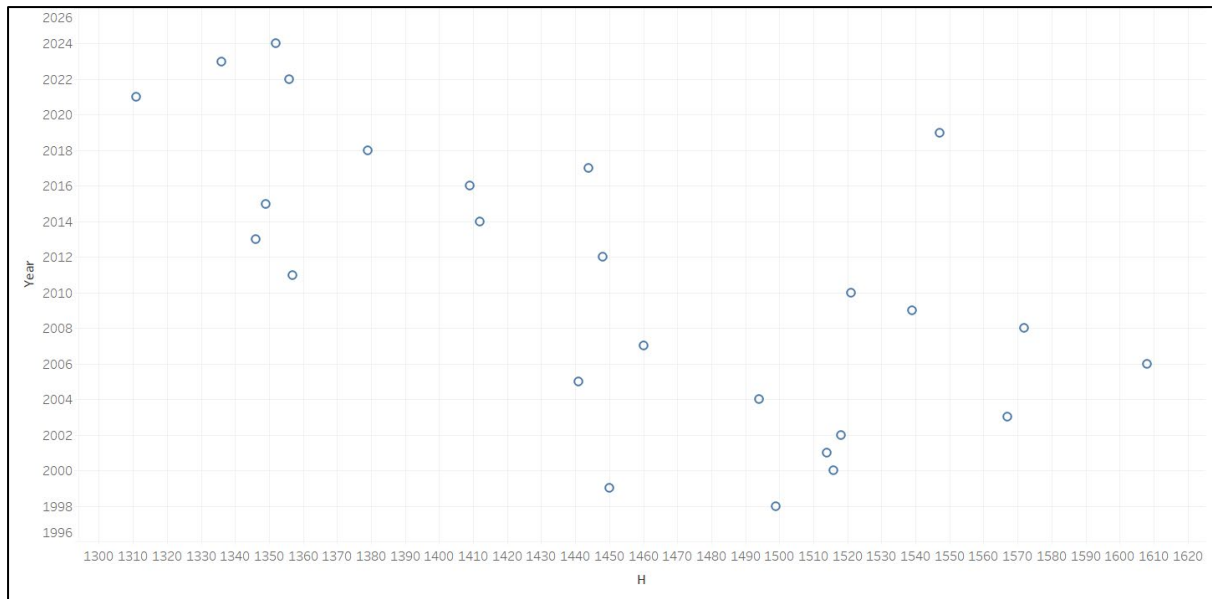
The distribution for walks (BB) seems to be normally distributed and 34.62% of total walks lie between 490 and 516. There are no potential outliers in the distribution.



The distribution for hits doesn't have any potential outliers and 26.92% of total hits lie between 1335 and 1380.

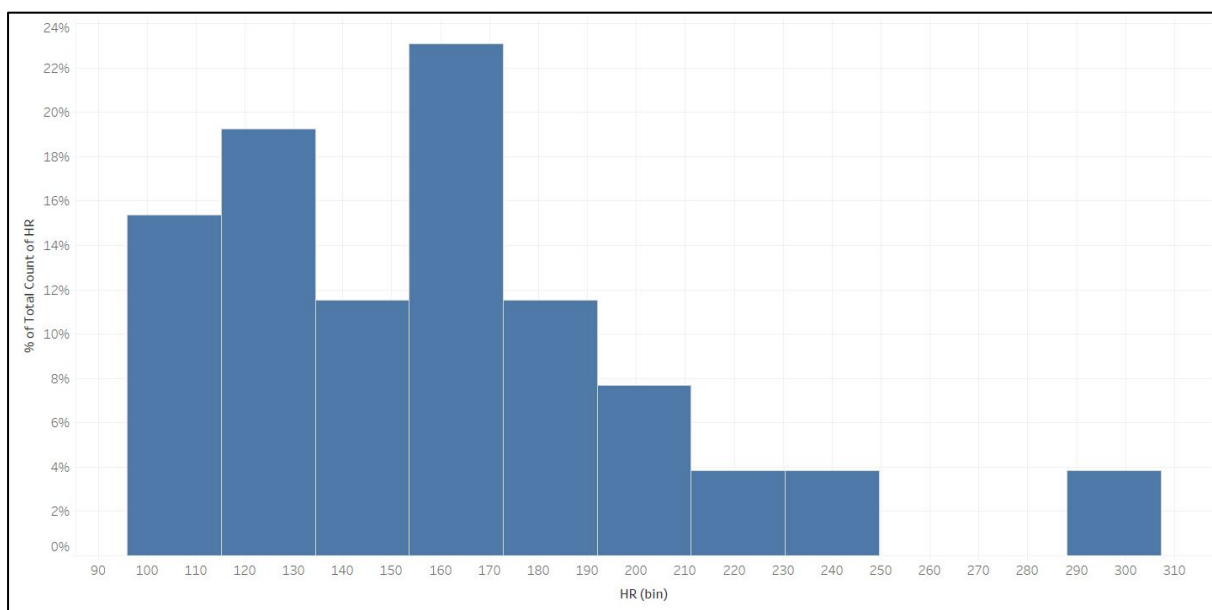


The highest number of hits (1608) was in 2006 and the lowest was (1311) in 2021.



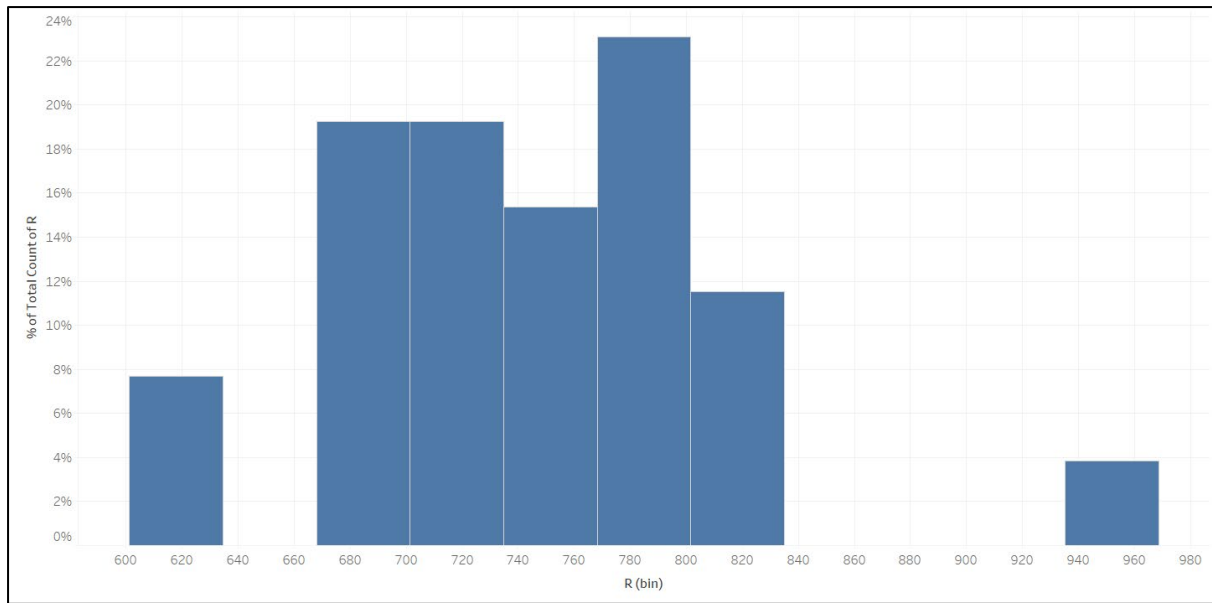
The distribution for home runs is right-skewed and there is one potential outlier.

23.08% of the total home runs lie between 154 and 172 and more than 50% (69.23%) of the total home runs lie between 96 and 172.

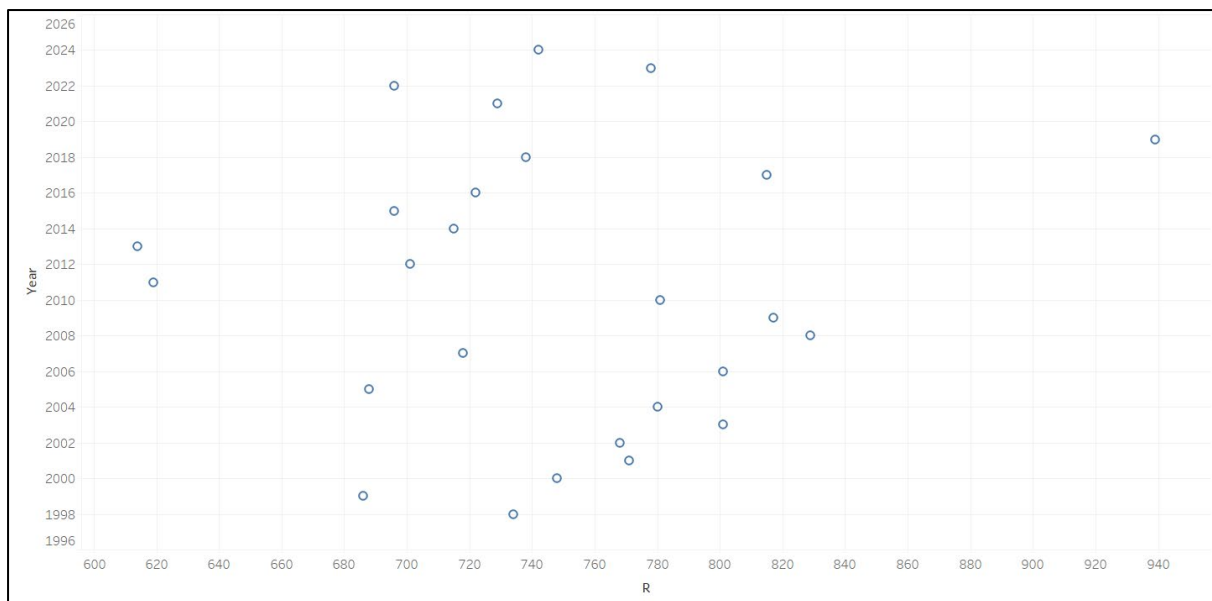


The distribution for runs is kind of normally distributed with some potential outliers.

23.08% of the total runs lie between 768 and 801.

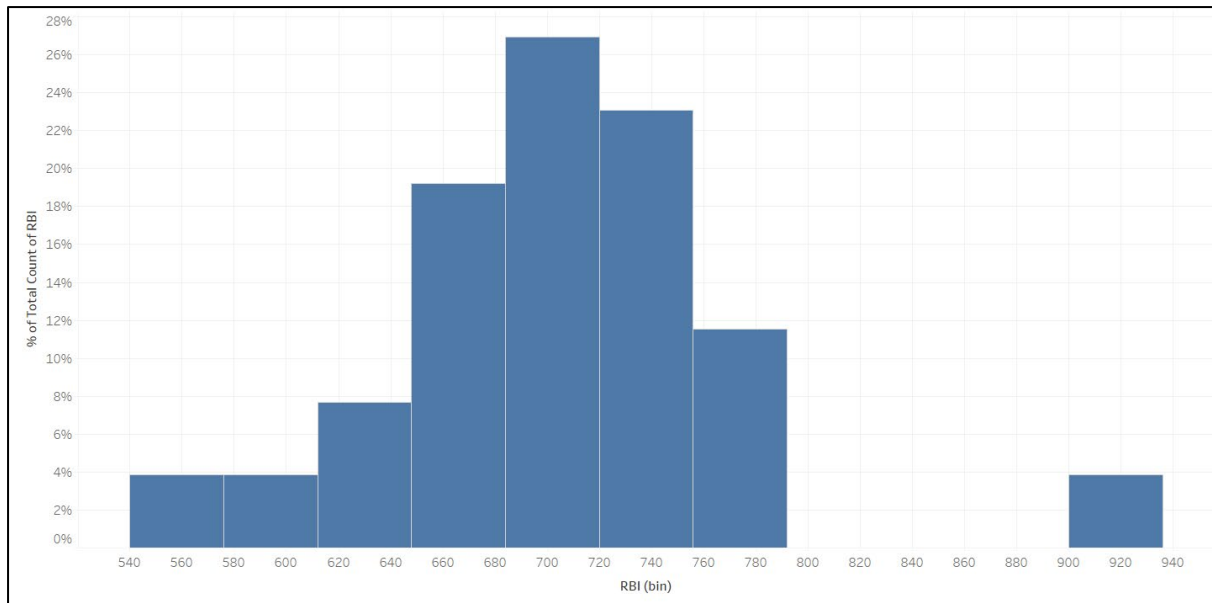


The lowest number of runs scored is 614 in year 2013 and the highest is 939 in year 2019.

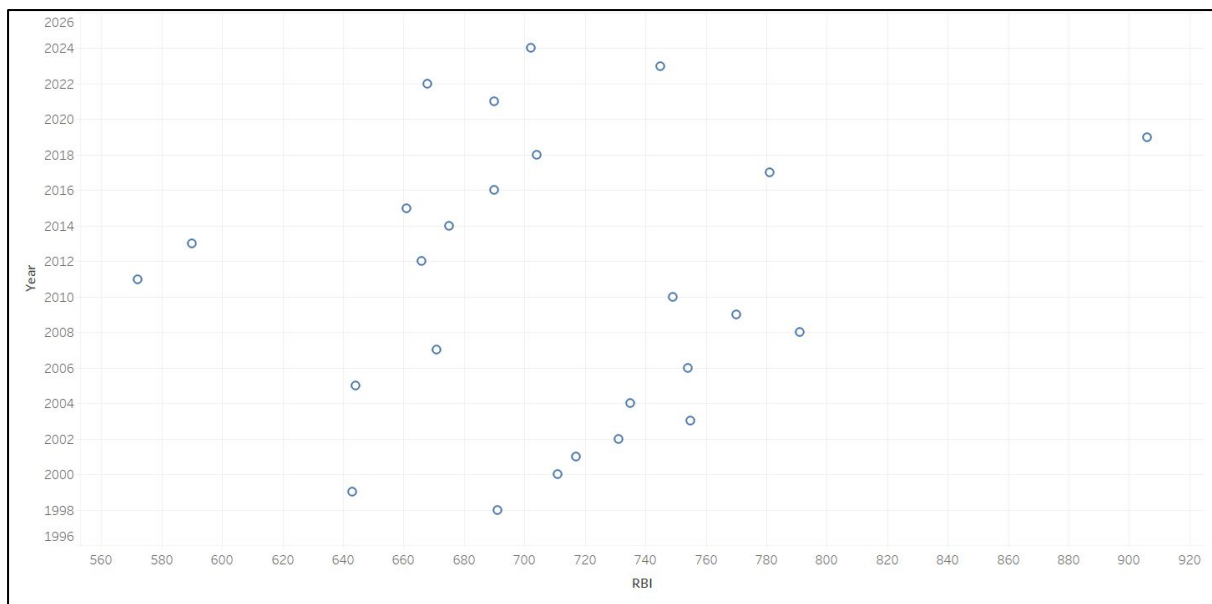


The distribution of Runs batted in is left-skewed with a potential outlier.

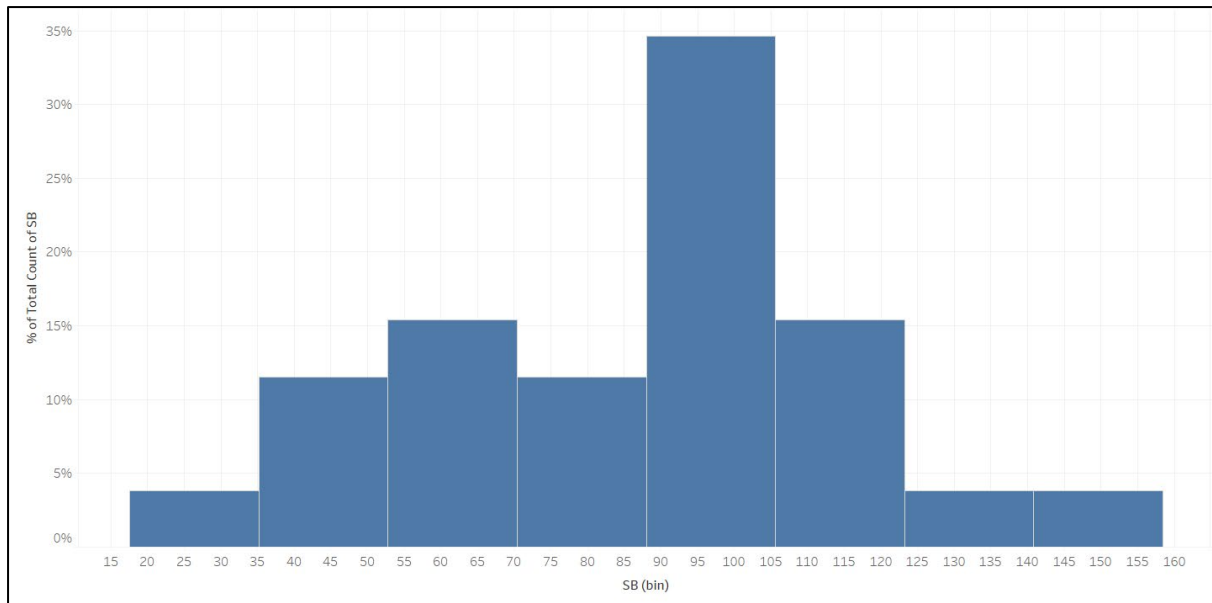
26.92% of the total runs batted in are between 684 and 720.



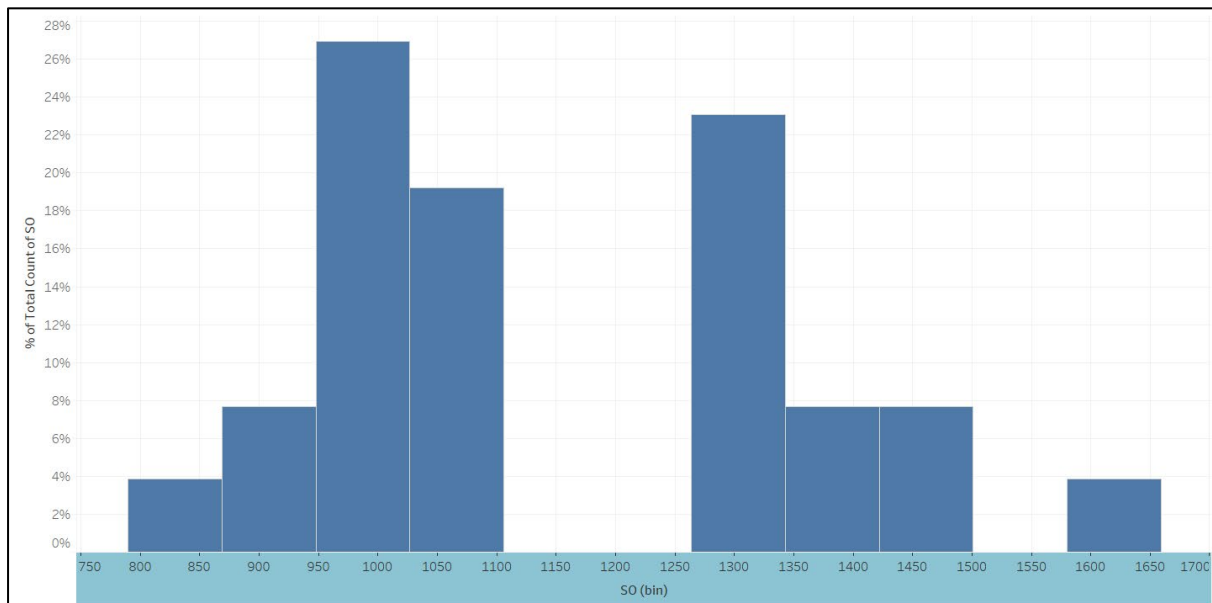
The lowest number of runs batted in are 572 in year 2011 and the highest are 906 in 2019.



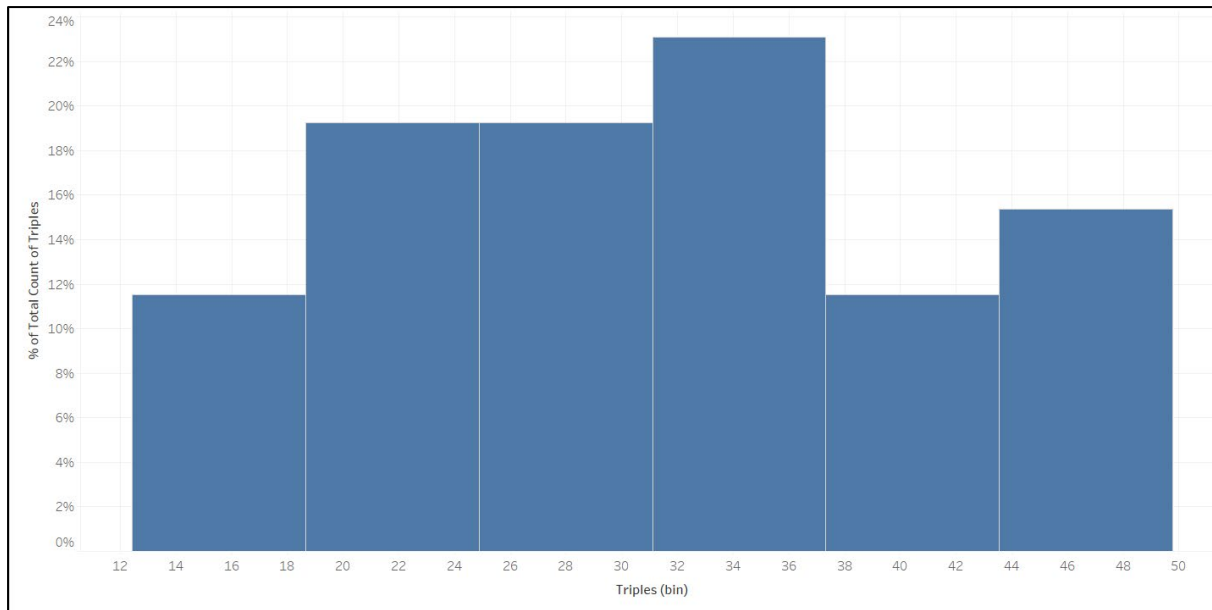
Stolen bases seems normally distributed with no potential outliers, 34.62% of the total stolen bases are between 88 and 105.



The distribution of strikeouts doesn't seem like a continuous one, with no strike outs between 1106 and 1264. There are potential outliers. 26.92% of the total strikeouts lie between 948 and 1027.



The distribution of triples seems to be normally distributed with no potential outliers. 23.08% of the total triples are between 31 and 36.



Methods

The analysis was conducted using various SAS procedures to examine relationships and predict team wins. For correlation analysis, the PROC CORR procedure was employed to quantify associations between variables. To visually assess relationships between metrics, the PROC SGPLOT procedure generated scatter plots, providing a clear view of patterns and potential trends. Interaction effects were evaluated using both PROC GLM and PROC PLM, enabling the exploration of how variables might jointly impact team wins. Finally, PROC REG was used to perform multiple regression analysis, identifying the best combination of predictors with statistical significance. These procedures together allowed for a comprehensive approach to analyzing and modeling the data.

Correlation Analysis

We will now conduct a correlation analysis to identify significant relationships between the independent variables and the dependent variable. The goal is to determine whether any correlations are strong enough to justify using the independent variables as predictors in a regression analysis. Pearson's correlation coefficient (r) will be used to measure these associations.

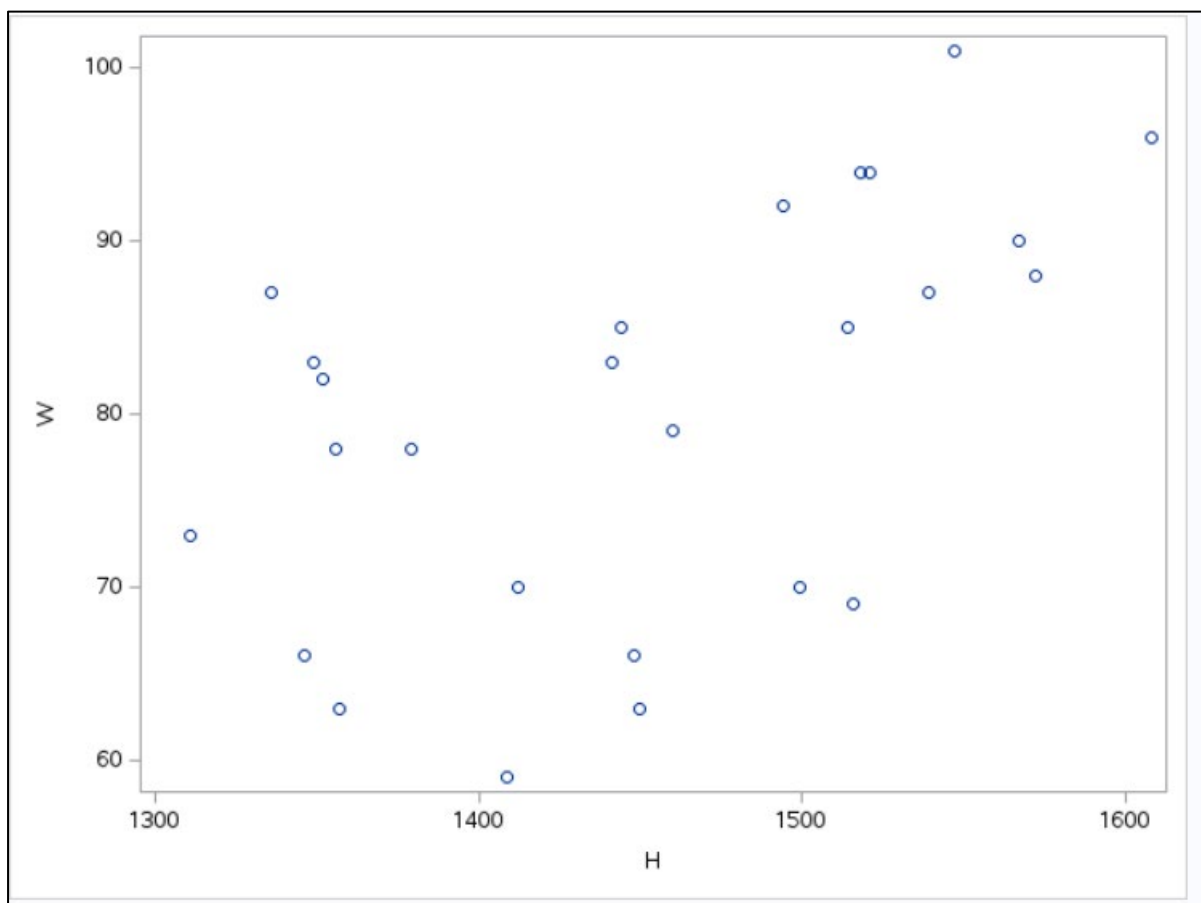
I will now assess the relationships between each one of the potential independent variables: hits(H), runs(R), doubles, batting average (BA), runs batted in (RBI), home runs (HR), triples, strikeouts (SO), walks (BB), age of the batters (BatAge) and stolen bases (SB) and the dependent variable, wins. The significance of the correlation can be evaluated by examining the p-value. If the p-value is less than the alpha level (0.05), we will reject the null hypothesis—which indicates that there is no correlation between the two variables. Conversely, if the p-value exceeds 0.05, we will fail to reject the null hypothesis, indicating that the correlation is not significant. If a significant correlation is found, we will then evaluate the strength of the relationship using the r value.

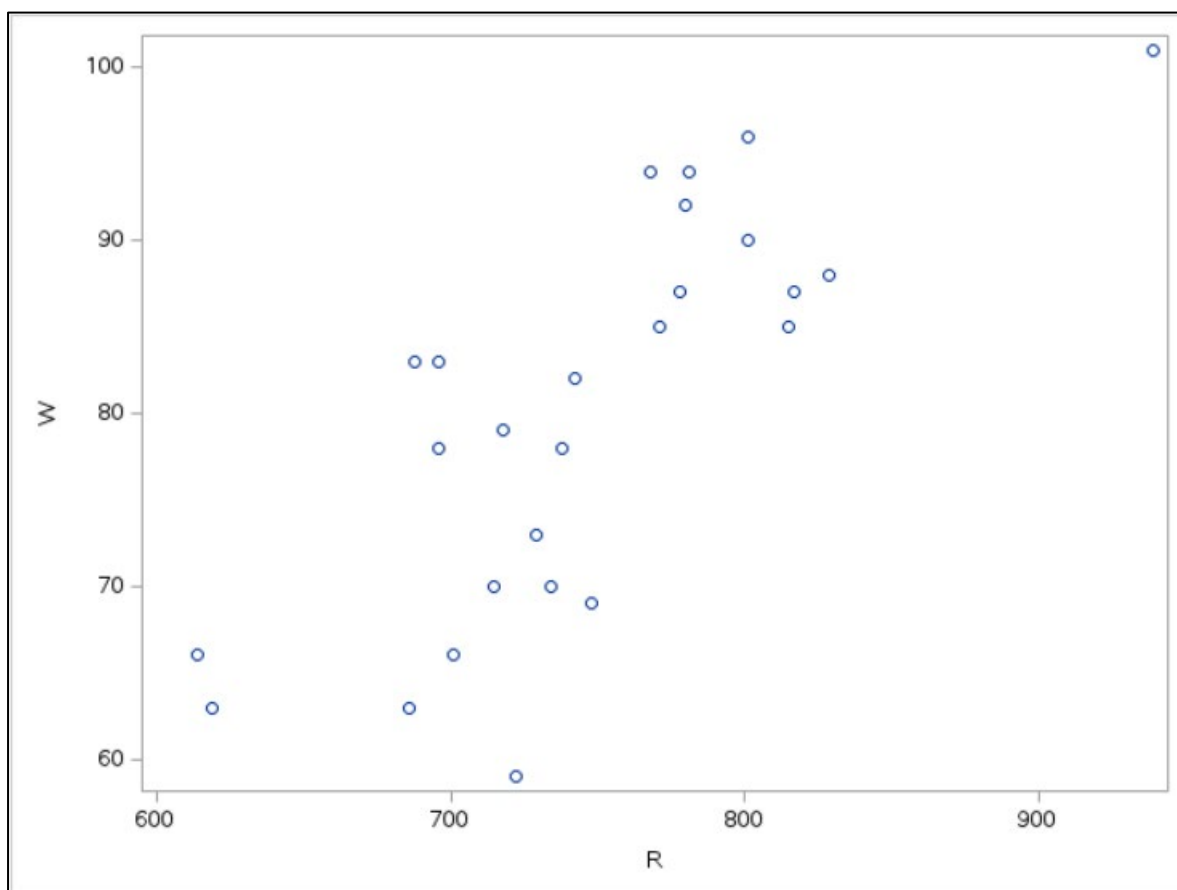
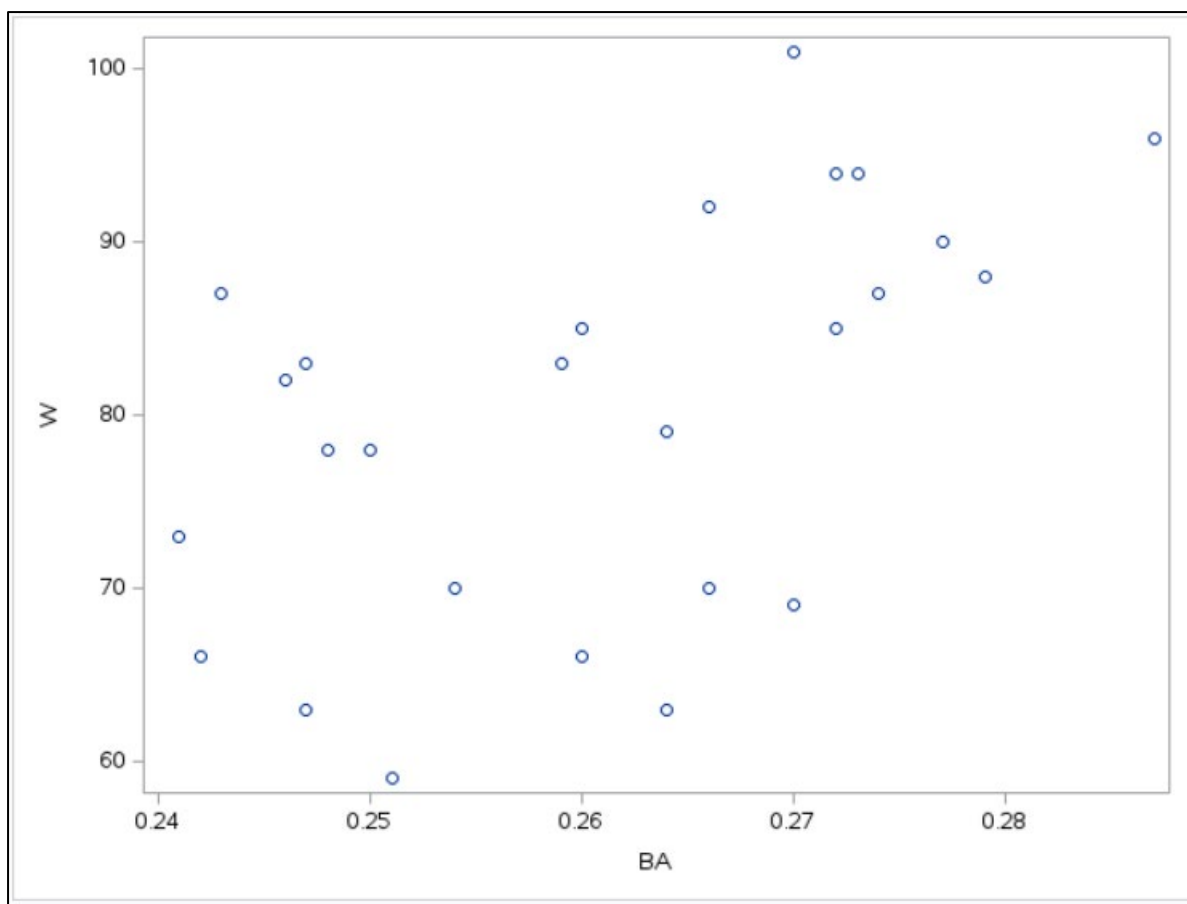
The correlation coefficient ranges from -1 to 1. An r of -1 or 1 indicates a perfect correlation. An r value between 0.7 and 0.9 or between -0.7 and -0.9 signifies a strong relationship, while an r -value between 0.4 and 0.7 or between -0.4 and -0.7 indicates a moderately strong relationship. An r between 0.1 and 0.4 or between -0.1 and -0.4 reflects a weak relationship, and if r is close to 0, it suggests little to no linear relationship. A positive sign indicates that as one variable increases, the other also increases, whereas a negative sign suggests that as one variable increases, the other decreases.

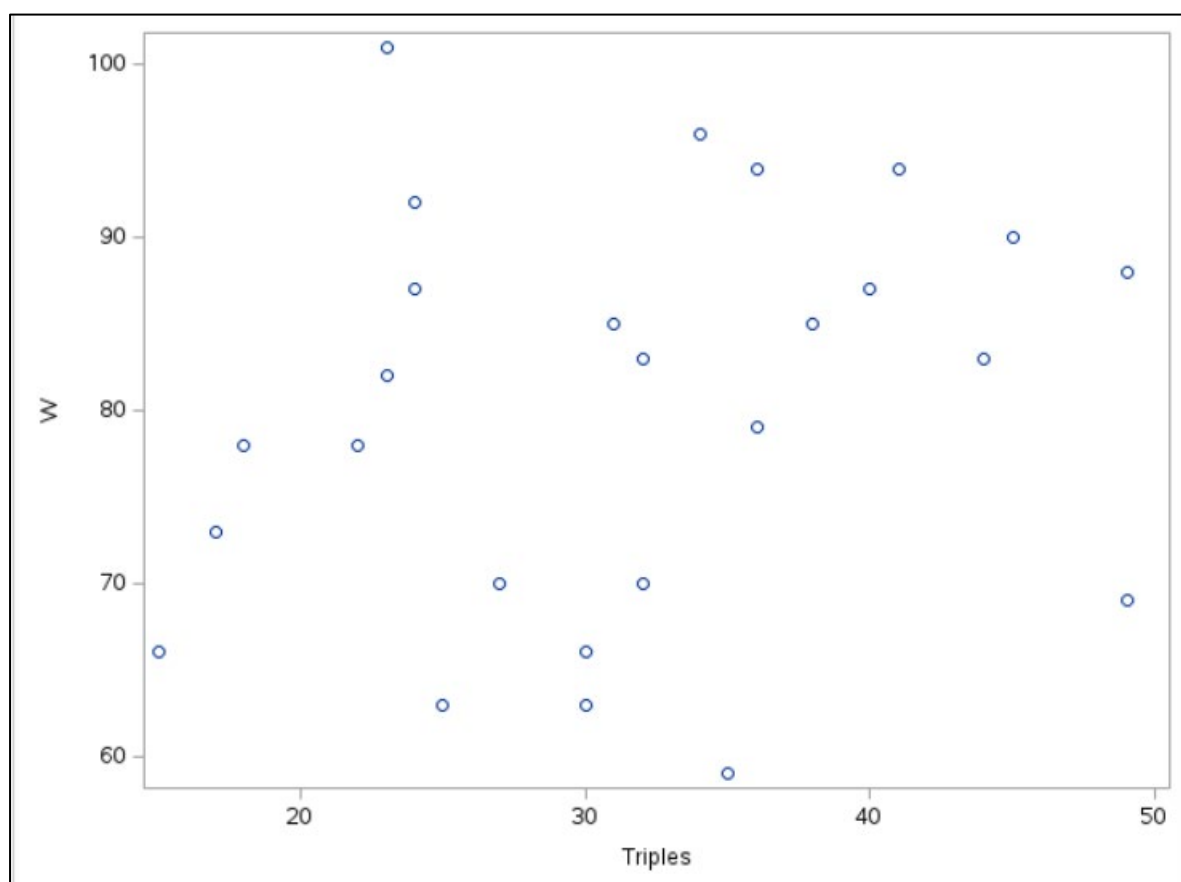
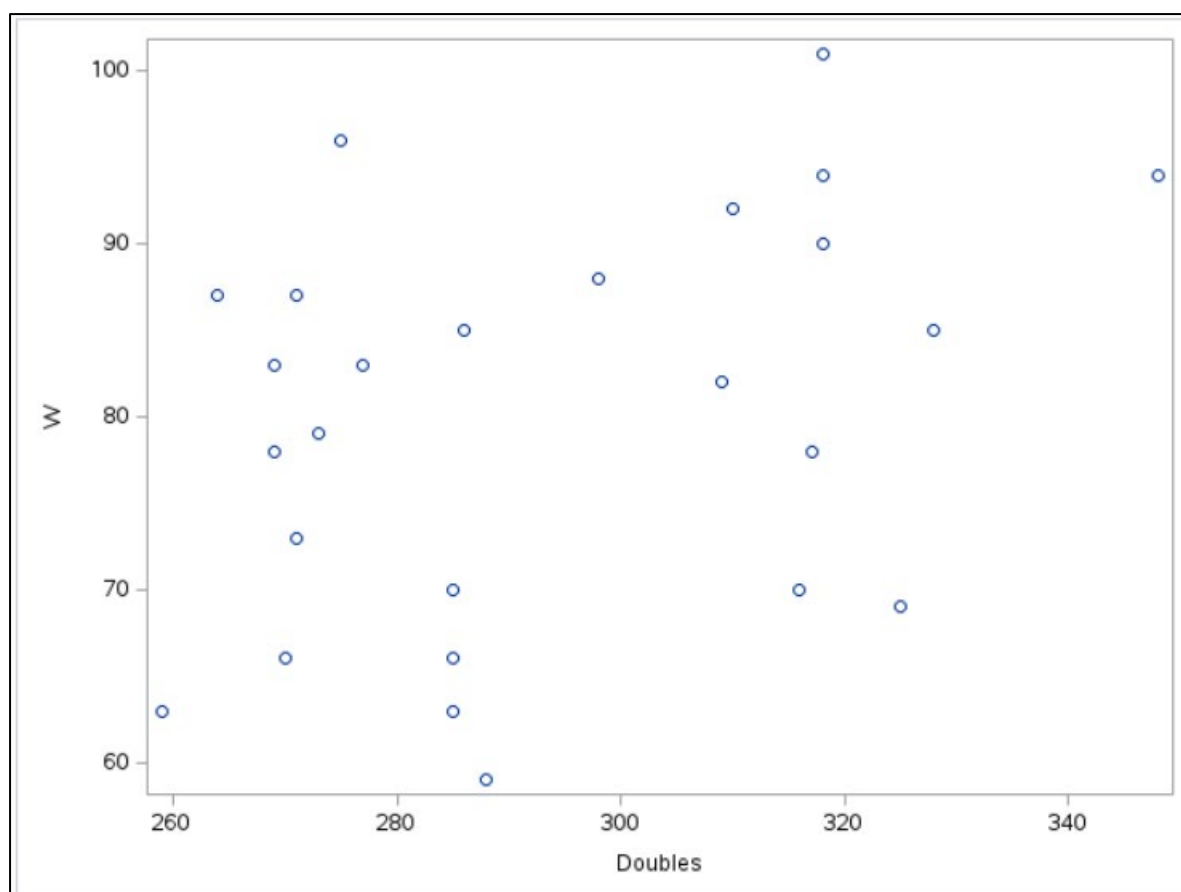
Among the independent variables, the p-values for the correlations between doubles and wins, triples and wins, stolen bases and wins, walks and wins, strikeouts and wins as well

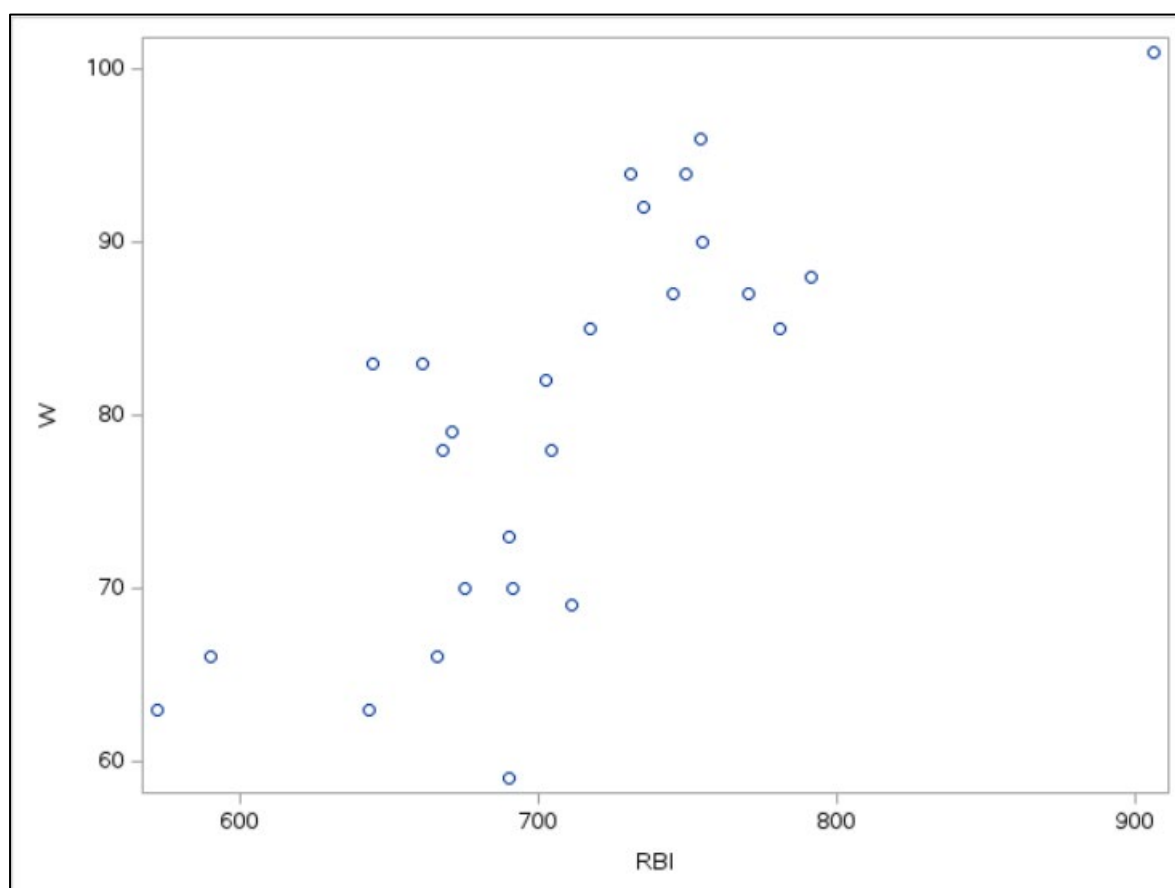
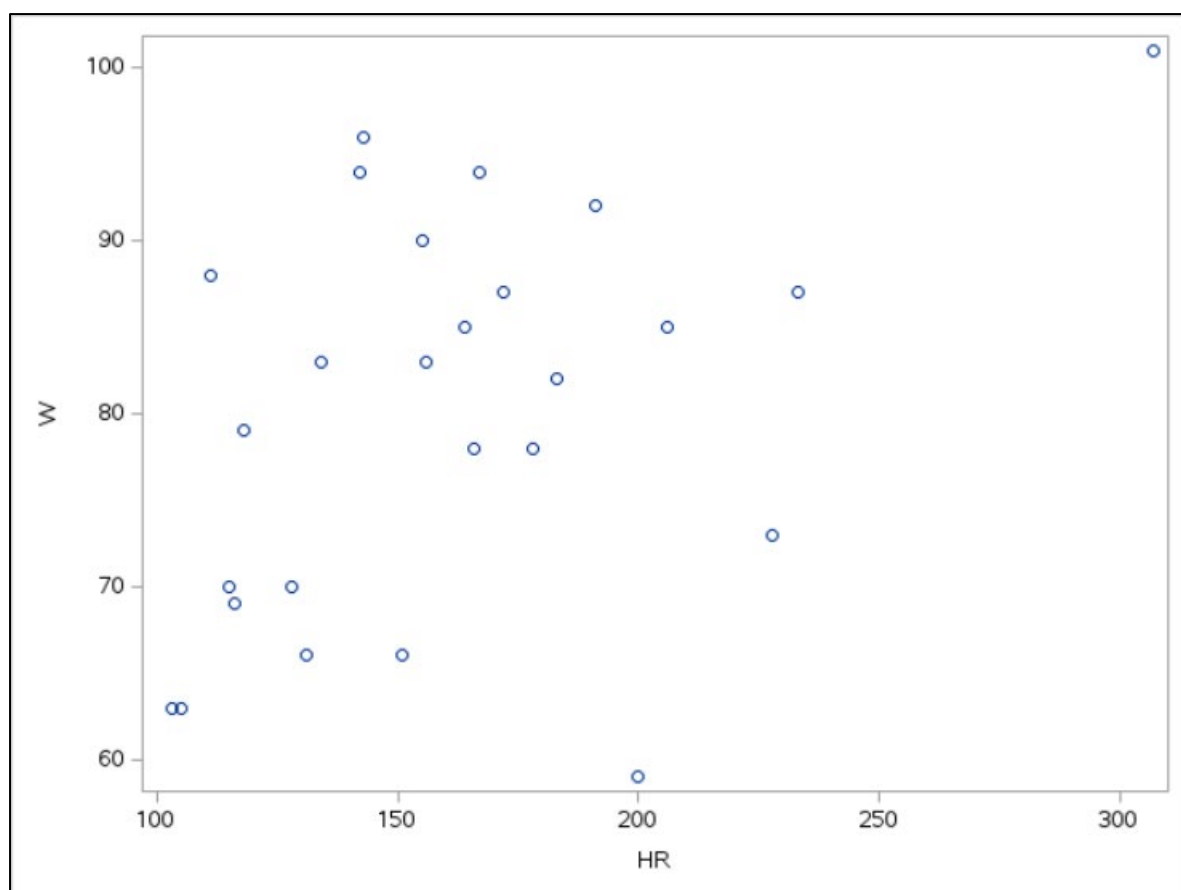
as the age of batters are greater than 0.05 ($p = 0.0782$, $p = 0.2835$, $p = 0.3837$, $p = 0.5262$, $p = 0.4578$ and $p = 0.8159$ respectively with r values of 0.35158, 0.21852, -0.17823, 0.13019, -0.155223 and -0.04799 respectively). This means we fail to reject the null hypothesis for these variables, indicating no correlation with wins. All other correlations show significant p -values ($p < 0.05$), allowing us to reject the null hypothesis and conclude that there is a relationship between the other independent variables and wins.

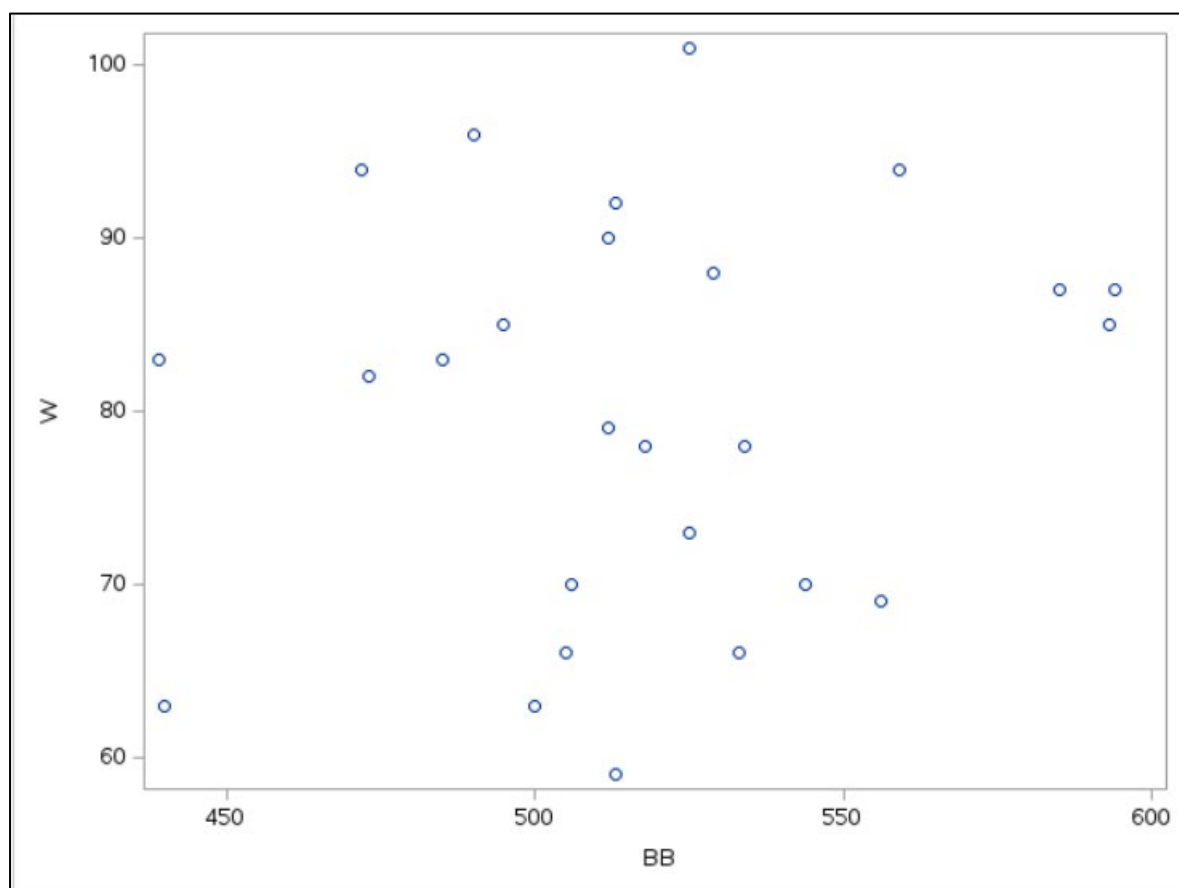
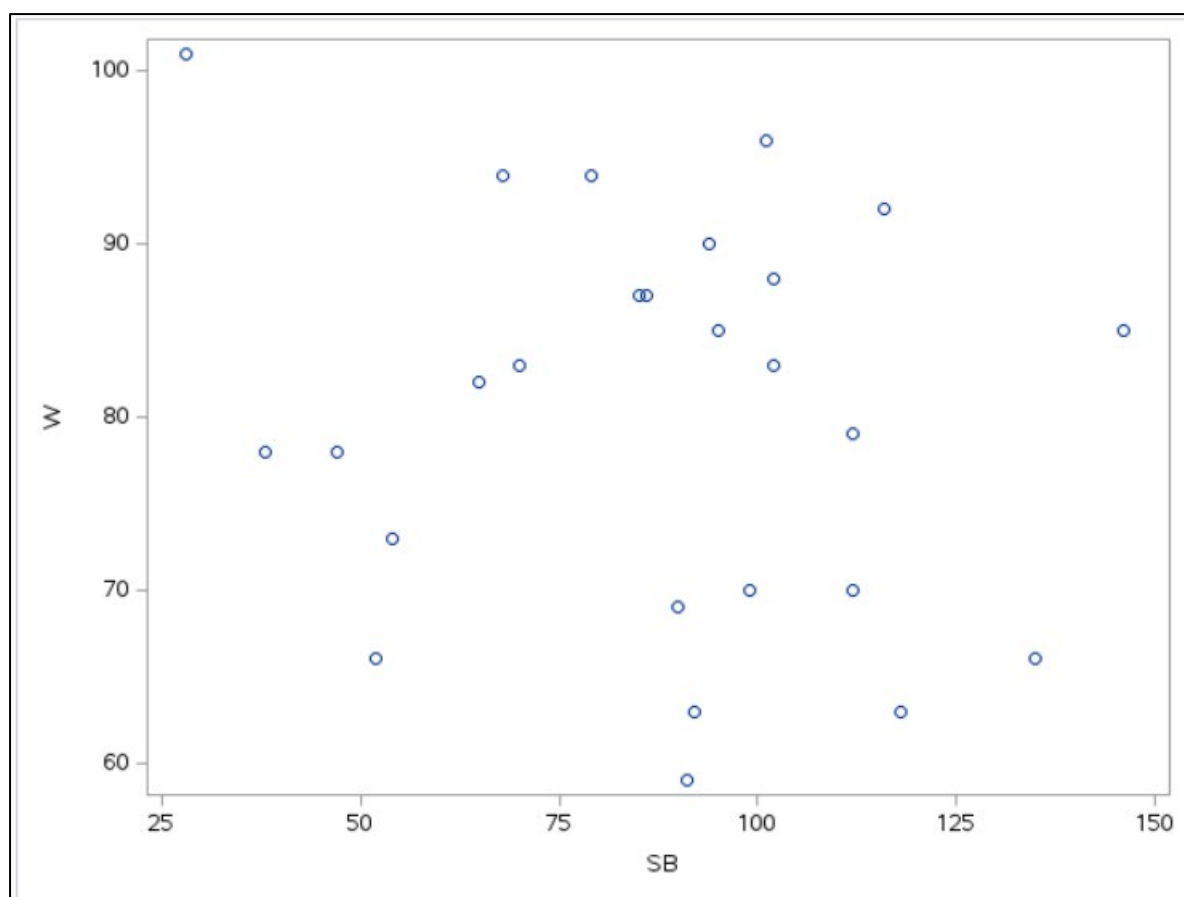
The strength of these relationships can be determined from the r values: there is a strong relationship between runs and wins ($r = 0.76611$) and runs batted in and wins ($r = 0.75527$), and a moderate relationship exists between hits and runs ($r = 0.53214$) and batting average and runs ($r = 0.53641$) as well as home runs and runs ($r = 0.42401$). Here are scatterplots illustrating these relationships.

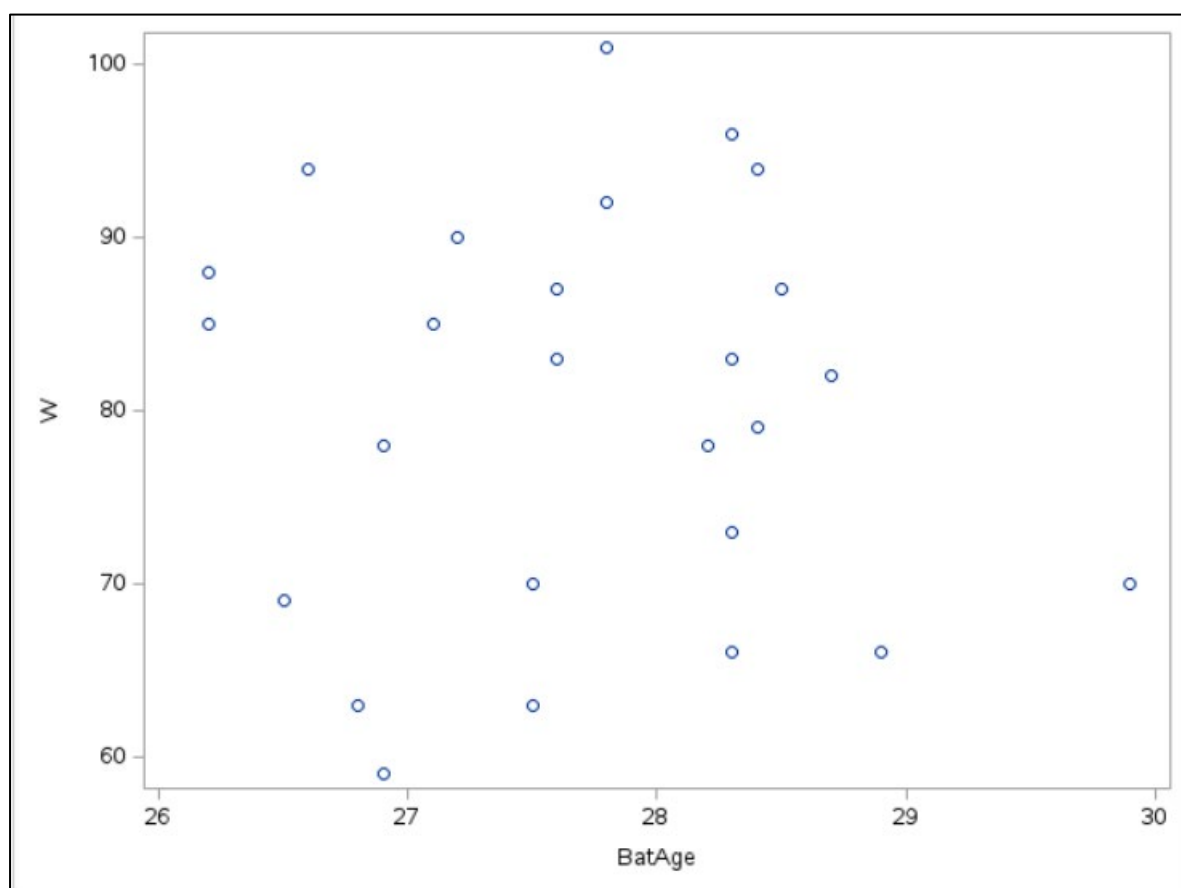
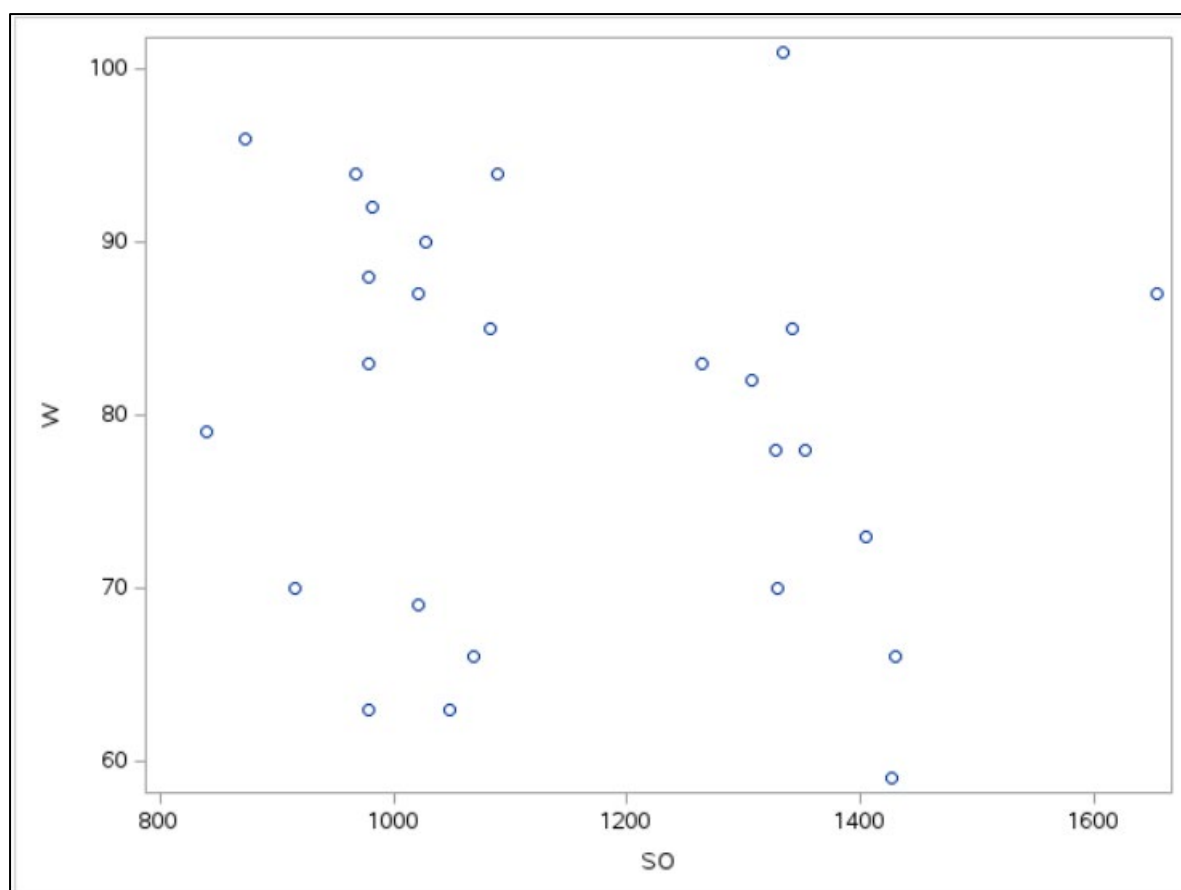












Runs batted in, runs, hits, and batting average are the most strongly associated with wins, making them our final four independent variables.

Multiple Linear Regression

We can now construct a series of two-variable models to identify the best one for predicting team wins. With four independent variables available, we can create six such models, each using wins as the dependent variable. These model combinations are: runs and hits, runs and runs batted in, runs and batting average, hits and runs batted in, hits and batting average, and runs batted in and batting average.

To assess model adequacy, the p-value of the F-statistic is examined. If this p-value exceeds 0.05, we fail to reject the null hypothesis that all independent variable slopes are zero ($\beta_1 = \beta_2 = 0$), suggesting that the model is inadequate. Conversely, if the p-value is below 0.05, we reject the null hypothesis, concluding that at least one variable's slope is non-zero, indicating an adequate model.

For evaluating model utility, we focus on several indicators, including individual t-test p-values for variable slopes, the adjusted R-squared, the F-statistic and its p-value, as well as the coefficient of variation. If an independent variable's t-test p-value is above 0.05, we fail to reject the null hypothesis that the variable's slope is zero, suggesting no significant association with the dependent variable. A p-value below 0.05 leads us to reject this null hypothesis, indicating a meaningful association. Ideally, we aim for as many p-values below 0.05 as possible; a model with all t-test p-values above 0.05 is generally considered weak. An adjusted R-squared closer to 1 indicates a better model fit, as does a higher F-statistic. A coefficient of variation below 10% is also preferable for a robust model. The following table presents these key statistics, facilitating model comparisons.

Table 1.1

Model	p-value			r-squared adj	F-statistic		Coefficient of Variation
	intercept	Slope1	Slope2		value	p	
Runs-hits	0.3599	0.004	0.7365	0.5533	16.8	<.0001	9.67678
Runs-runs batted in	0.2624	0.2918	0.6312	0.5556	16.63	<.0001	9.65162
Runs-batting average	0.2722	0.0004	0.5159	0.5594	16.87	<.0001	9.61059
Hits-runs batted in	0.3415	0.5081	0.0005	0.5421	15.8	<.0001	9.76710
Hits-batting average	0.4652	0.9499	0.7007	0.2259	4.65	0.0202	12.73771
Runs batted in -batting avg	0.2305	0.0004	0.3274	0.5526	16.44	<.0001	9.68433

From the table, the models with runs-runs batted in and hits-batting average don't even have one p-value that is less than 0.05 for the individual t-tests so, they are not good models. All the other models have one p-value that is less than 0.05 for the individual t-tests and among these models, the model with runs-batting average has the highest r-squared adjusted value, F-statistic, and the least coefficient of variation. So, the runs-batting average model is the best one among the 6 models.

Now, we will add one of the two remaining of the four independent variables to this two-variable model and create a three-variable model. We can create 2 three-variable models, here is a table with the relevant statistics to compare the two models,

Table 1.2:

Model	p-value				r-squared adj	F-statistic		Coefficient of Variation
	intercept	Slope1	Slope2	Slope3		value	p	
Runs-batting average-hits	0.0470	0.0001	0.0725	0.0870	0.5980	13.39	<.0001	9.17981

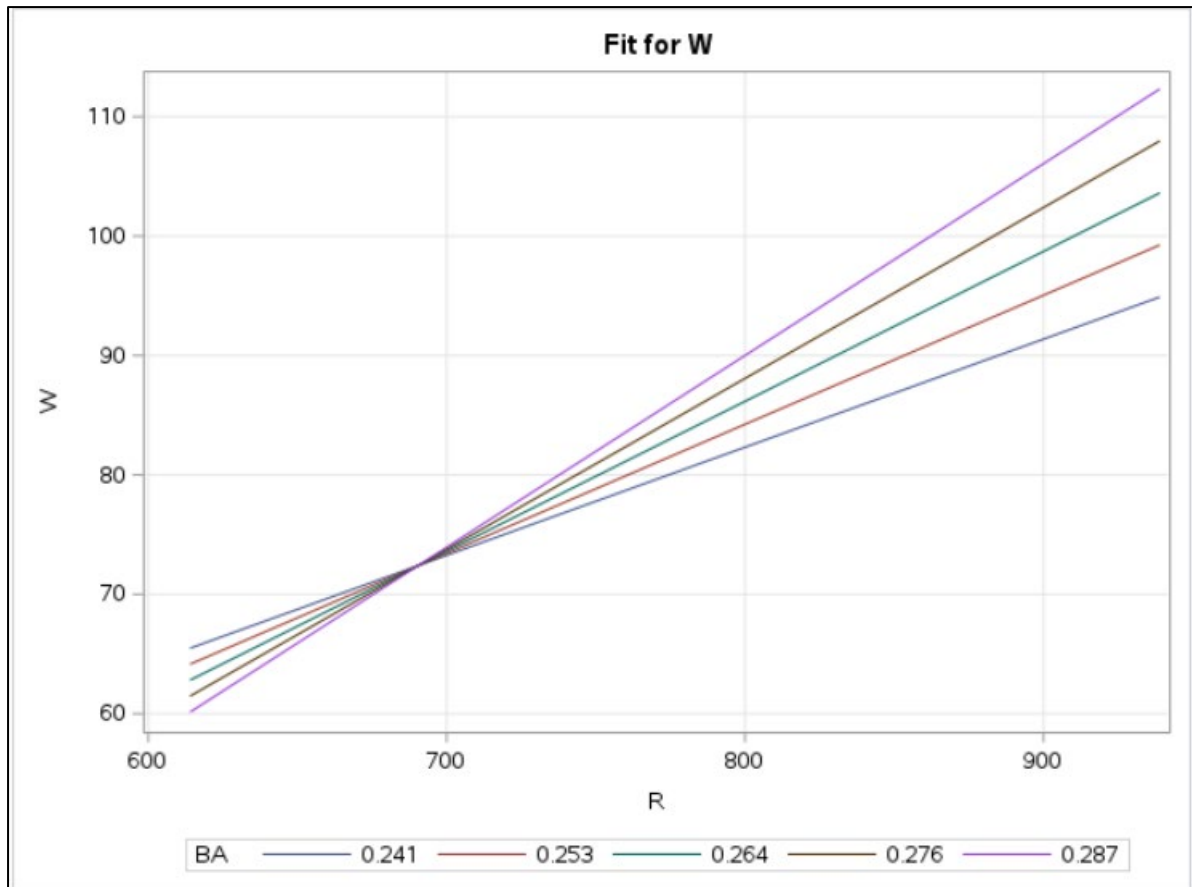
Runs-batting average-runs batted in	0.3147	0.5504	0.6459	0.8674	0.5399	10.78	<.0001	9.82022
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From the table, the model runs-batting average-runs batted-in model doesn't even have one p-value less than 0.05 for the individual t-tests so, the runs-batting average-hits model is the best one among the two.

However, this three-variable model doesn't seem to be adding much value compared to the two-variable model (runs-batting average). Even though the r-squared adjusted is slightly higher for the three-variable model, it isn't a large difference so we can choose the parsimonious model, the one with fewer variables. We can also check this by doing an F-test for the nested models, with the three-variable model (runs-batting average-hits) as the complete model and the two-variable model (runs-batting average) as the reduced model. The value of the F-statistic is 3.21 and the p-value is 0.0870 which is higher than 0.05, so we fail to reject the null hypothesis (slope of hits = 0), concluding that the extra variable (hits) doesn't contribute information to the prediction of the wins of the team. So, our best model until now is the runs-batting average model.

From Table 1.1, we can see that the t-test of the slope of the batting average is not significant, so we can try adding an interaction term or second-order terms to get a much better model.

We can create an interaction plot to determine if an interaction term (runs*batting average) for this model is required. Here is the plot,



From the picture, we can see that an interaction term might be required as the lines all converge at one point.

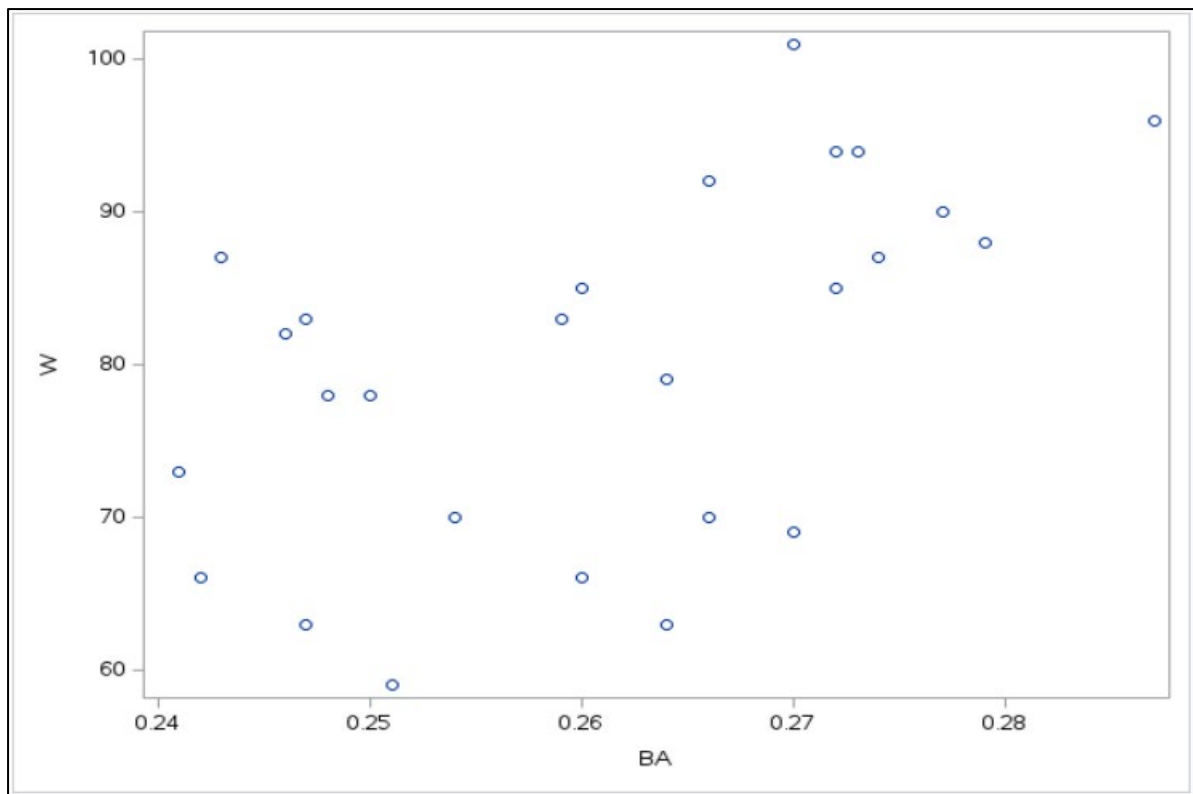
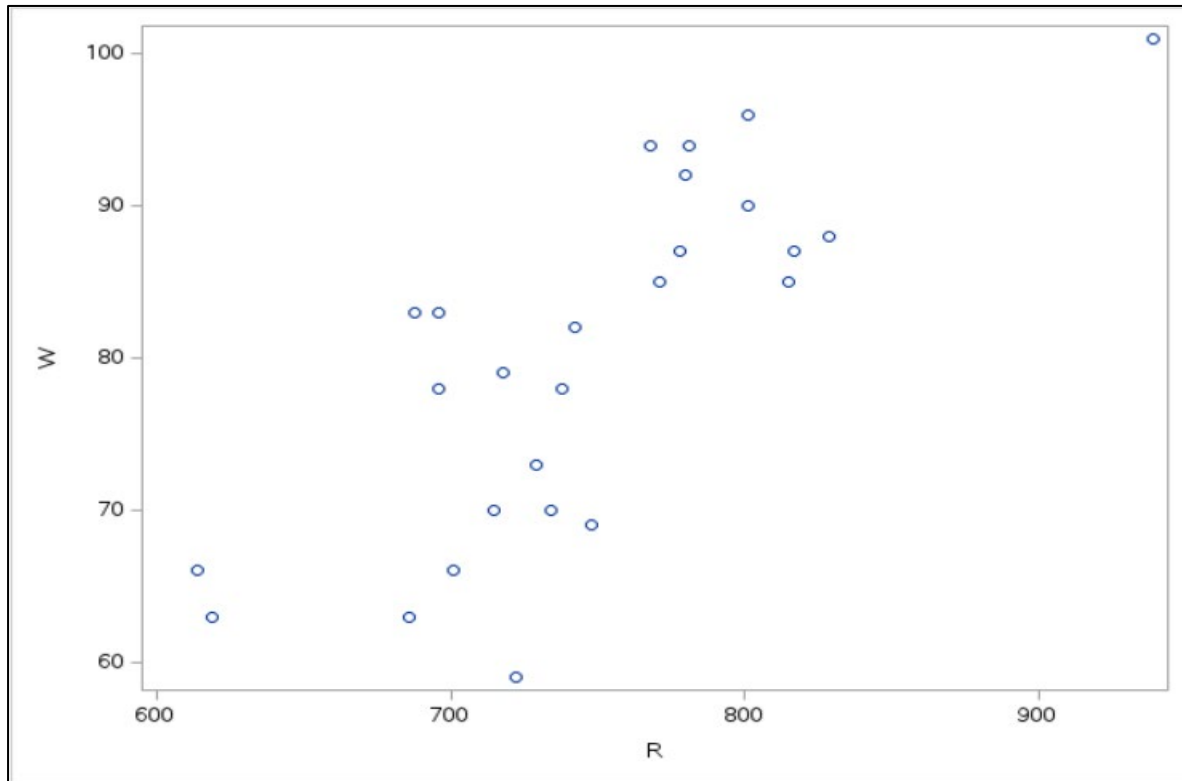
Here is the table of all the relevant statistics for the interaction model (runs-batting average- runs*batting average),

Table 1.3:

Model	p-value				r-squared adj	F-statistic		Coefficient of Variation
	intercept	Slope1	Slope2	Slope3		value	p	
Interaction model	0.5380	0.6250	0.5244	0.4847	0.5497	11.17	0.0001	9.71567

The p-value of the interaction term is greater than 0.05 so, we fail to reject the null hypothesis (slope of the interaction term (β_3) = 0) concluding that the interaction term is not statistically significant and does not add value to the model.

We can look at the scatter plots of the variables in the runs-batting average model i.e., runs and batting average to determine if their second-order terms (runs*runs, batting average*batting average) are required. Here are the scatter plots,



The relationship between wins and batting average doesn't exactly seem linear so we can try adding the second-order term of batting average to the model. We can also create 3 models, two with second-order terms of only one of the variables and one with both the second terms. Here is the comparison table,

Table 1.4:

Model	p-value				r-squared adj	F-statistic		Coefficient of Variation
	intercept	Slope1	Slope2	Slope3		value	p	
Runs-batting average – runs*runs	0.6790	0.6218	0.5488	0.8988	0.5397	10.77	0.0001	9.82289
Runs-batting avg- batting avg*batting avg	0.1789	0.0004	0.1686	0.1628	0.5792	12.47	<.0001	9.39163

Model	p-value					r-square d adj	F-statistic		Coefficient t of Variation
	intercept	Slope1	Slope2	Slope3	Slope4		val	p	
Rus-batting avg- runs*runs- batting avg*batting avg	0.2002	0.5821	0.1765	0.8539	0.1708	0.5599	8.95	0.0002	9.60470

From the tables, the models runs-batting average-runs*runs and runs-batting average-run*runs-batting average*batting average don't even have one p-value less than 0.05 for the individual t-tests so they are not good models. So, the runs-batting avg-batting avg*batting avg is the best model among the three but the p-value of the second-order term (batting-batting avg) is greater than 0.05, so we fail to reject the null hypothesis (slope of the second-

order term ($\beta_3 = 0$) concluding that the second order term is not statistically significant and does not add value to the model.

We can try adding both the interaction term and the second-order term of batting average to see if the model (Runs-batting average- batting avg*batting avg-runs*batting avg) is better than the two-variable model (runs-batting average). Here is the table with relevant statistics,

Table 1.5:

Model	p-value					r-square d adj	F-statistic		Coefficient of Variation
	intercept	Slope1	Slope2	Slope3	Slope4		val	p	
	0.1932	0.8573	0.1894	0.2382	0.9992	0.5592	8.93	0.0002	9.61264

From the table, none of the individual t-tests have a p-value less than 0.05, so this model is not good.

So, we can conclude that the two-variable model (runs-batting average) is the best model to predict the team wins. The linear model for wins, runs and batting average from the time period 1998-2024, excluding 2020 is (with an alpha value of 0.05),

$$\text{Wins} = -35.02929 + (0.11946) * (\text{Runs}) + (98.94998) * (\text{Batting Average})$$

We can interpret this as, if the batting average and runs are zero, then the number of wins would be -35.02929, if runs are increased by 1 unit then, the number of wins increases by 0.11946 units while the batting average is kept fixed and if the batting average is increased by 1 unit then wins increases by 98.94998 units while runs are kept constant.

The sum of squares error (SSE) is 1360.89859, standard deviation (s) and variance (s^2) are 7.69217 and 59.16950 respectively. We expect most (approximately 95%) of the observed values of wins to lie within 15.38434 units of the least squares regression line.

The p-value of the F-statistic (16.87) is $<.0001$ which is less than 0.05, we reject the null hypothesis (slope of runs = slope of batting average = 0), concluding that there slope of at least one of the variables is not zero. The p-value of runs is less than 0.05 (0.0004), we reject the null hypothesis (slope of runs = 0), concluding that there is a relationship between wins and runs but the p-value of batting average is greater than 0.05 (0.5159), we fail to reject the null hypothesis (slope of batting average = 0), concluding that there is no relationship between wins and batting average. The value of r-squared adjusted is 0.5594, which means 55.94% of the variance in number of wins can be explained by the model.

Final Conclusion and Next Steps

In conclusion, the two-variable model using runs and batting average was determined to be the optimal predictor of team wins. This model explained 55.94% of the variance in wins, indicating that runs are significantly correlated with wins, while batting average, despite being included, was not statistically significant in predicting wins alone. This model indicates that for every unit increase in runs, the number of wins is expected to increase by 0.11946, holding the batting average constant. Similarly, a one-unit increase in batting average corresponds to an increase of 98.94998 wins, assuming runs remain fixed. Further validation of this model with additional seasons or adjusted data may provide more robust insights into its predictive accuracy.

I would explore the influence of additional variables to evaluate their impact on team wins. I'm also interested in understanding how well a model using strikeouts and hits as predictors might perform in estimating team wins. Future analyses could consider individual

player performances to identify key contributors to wins, as well as examine contextual factors like the coaching staff, game location, and opponent teams. This approach would offer a deeper understanding of the dynamics at play and might highlight specific conditions or team matchups that influence game outcomes.

Appendix:

Code used:

To get data for years from 1998 to 2024:

```
Data twins2;  
    set twins;  
    if year >= 1998;  
run;
```

For Correlation:

```
proc corr data = twins2;  
    Var W doubles;  
run;
```

```
proc corr data = twins2;  
    Var W H;  
run;
```

```
proc corr data = twins2;  
    Var W BA;  
run;
```

```
proc corr data = twins2;
```

```
Var W R;
```

```
run;
```

```
proc corr data = twins2;
```

```
Var W triples;
```

```
run;
```

```
proc corr data = twins2;
```

```
Var W R;
```

```
run;
```

```
proc corr data = twins2;
```

```
Var W RBI;
```

```
run;
```

```
proc corr data = twins2;
```

```
Var W HR;
```

```
run;
```

```
proc corr data = twins2;
```

```
Var W SB;
```

```
run;
```

```
proc corr data = twins2;
```

```
Var W BB;
```

```
run;
```

```
proc corr data = twins2;
```

```
Var W SO;
```

```
run;
```

```
proc corr data = twins2;
```

```
Var W BatAge;
```

```
run;
```

For scatter plot:

```
proc sgplot data = twins2;
```

```
scatter y = W x = H;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = BA;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = R;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = doubles;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = triples;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = HR;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = RBI;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = SB;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = BB;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = SO;
```

```
run;
```

```
proc sgplot data = twins2;
```

```
scatter y = W x = Batage;
```

```
run;
```

For regression:

```
proc reg data = twins2;
```

```
model W = R H;
```

```
run;
```

```
proc reg data = twins2;
```

```
model W = R RBI;
```

```
run;
```

```
proc reg data = twins2;
```

```
model W = R BA;
```

```
run;
```

```
proc reg data = twins2;
```

```
model W = H RBI;
```

```
run;
```

```
proc reg data = twins2;
```

```
model W = H BA;
```

```
run;
```

```
proc reg data = twins2;
```

```
model W = RBI BA;
```

```
run;
```

```
proc reg data = twins2;
```

```
model W = R BA H;
```

```
test H;
```

```
run;
```

```
proc reg data = twins2;
```

```
model W = R BA RBI;
```

```
run;
```

For interaction plot:

```
proc GLM data = twins2;
```

```
model W = R | BA / solution;  
store GLMMODEL;
```

```
proc PLM restore = GLMMODEL noinfo;  
  effectplot slicefit(x = R sliceby = BA);  
run;
```

To add interaction term:

```
Data twins3;  
  set twins2;  
  int = R*BA;  
run;
```

For regression:

```
proc reg data = twins3;  
  model W = R BA int;  
run;
```

To include second order terms:

```
Data twins4;  
  set twins3;  
  sor = R*R;  
  sob = BA*BA;  
run;
```

For regression:

```
proc reg data = twins4;  
  model W = R BA sor sob;  
run;
```

```
proc reg data = twins4;
```

```
model W = R BA sor;  
run;
```

```
proc reg data = twins4;  
    model W = R BA sob;  
run;
```

```
proc reg data = twins4;  
    model W = R BA sob int;  
run;
```